

ATVB IN FOCUS: Cardiovascular Health in Women

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Placenta Biology, Future Cardiovascular Health, and Impact on the Maternal-Fetal Heart

Meenakshi Rana, Una Rose Wilcox¹, Suchita Nadkarni¹

ABSTRACT: The placenta is a highly specialized organ that ensures the developing fetus gets adequate oxygen and nutrients from its mother. The placenta shares common development pathways with the fetal heart and releases factors that can influence the way the mother's heart functions. Thus, pregnancy complications, where the placenta fails to develop properly, can lead to the development of fetal heart defects and induce long-term maternal cardiac dysfunction. In this review, we focus on how the placenta shapes cardiac events during development and the subsequent long-term impact on the offspring's heart. We also touch upon the potential role the placenta plays in maternal heart function.

Key Words: eclampsia ■ extravillous trophoblasts ■ fetal growth retardation ■ growth and development ■ hypertension, pregnancy-induced

The placenta ensures adequate fetal development while facilitating maternal physiological adaptation to pregnancy. Once placental structures, including villous trees, a cytotrophoblast shell, and extravillous trophoblasts invading the decidua, are established, they collectively support nutrient and gas exchange, waste removal, endocrine signaling, and immune regulation.¹

The multinucleated syncytiotrophoblasts facilitate the transfer of oxygen, glucose, amino acids, fatty acids, and micronutrients while simultaneously eliminating carbon dioxide and metabolic waste products.² The placenta is also a highly active endocrine organ that produces a range of hormones, shaping local and systemic maternal physiology while contributing to fetal growth and development through hormone secretion.^{3,4} Another key function of the placenta is immune regulation. Despite being semi-allogeneic, the fetus is tolerated by maternal immunity largely due to specialized mechanisms at the placental interface. Extravillous trophoblasts express nonclassical HLA molecules that engage uterine natural killer cells and modulate immune tolerance.^{5,6} At the same time, the syncytiotrophoblasts provide a physical and immunologic barrier against maternal immune attack and infection, producing antiviral type I and III interferons

and other cytokines that enhance fetal protection.⁷ This immunomodulatory environment ensures balance between tolerance to fetal antigens and defense against pathogens.

The role of the placenta in hypertensive disorders of pregnancy has been recently reviewed by our group.⁸ Briefly, in hypertensive disorders of pregnancy such as preeclampsia and eclampsia, abnormal trophoblast invasion, oxidative stress, and an imbalance in angiogenic mediators, including elevated sFlt-1 (soluble fms-like tyrosine kinase-1) and reduced PlGF (placental growth factor), compromise maternal vascular adaptation. These changes are clinically relevant, as the sFlt-1/PlGF ratio is now used to aid diagnosis and predict short-term adverse outcomes in preeclampsia.⁹ Placental pathology in these conditions contributes to intrauterine growth restriction (IUGR), preterm birth, and, in severe cases, stillbirth.¹⁰ Interestingly, chronic hypertension in pregnancy has been shown to be independently associated with placental dysfunction and increased placenta-mediated complications independent of preeclampsia, indicating early adverse effects on placental development and function.¹¹ Low-dose aspirin (75 mg/d) is currently given to reduce the risk of preeclampsia in women at

Correspondence to: Suchita Nadkarni, Centre for Biochemical Pharmacology, William Harvey Research Institute, Queen Mary University of London, London, United Kingdom. Email s.nadkarni@qmul.ac.uk

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Nonstandard Abbreviations and Acronyms

CHD	congenital heart disease
IUGR	intrauterine growth restriction
mTOR	mammalian target of rapamycin
NT-proBNP	N-terminal pro-B-type natriuretic peptide
PIGF	placental growth factor
sFlt-1	soluble fms-like tyrosine kinase-1
VEGF	vascular endothelial growth factor

high risk (risk factors include diabetes, chronic or gestational hypertension, renal disease, autoimmune disease, and previous history of preeclampsia) from ≤ 16 weeks of pregnancy.^{12,13} Aspirin inhibits the COX-1 enzyme and subsequent attenuation in thromboxane A2 (a vasoconstrictor), leading to an increase in prostacyclin (a vasodilator), the culmination of which leads to improved placental blood flow.¹⁴

Metabolic disorders are also known to impose significant stress on the placenta. In gestational diabetes, maternal hyperglycemia and insulin resistance drive structural remodeling, such as thickening of the vasculosyncytial membrane and altered villous architecture.¹⁵ These changes impair nutrient exchange and disturb signaling through insulin-IGF and mTOR (mammalian target of rapamycin) pathways, resulting in excessive fetal growth (macrosomia) or altered long-term metabolic programming.^{16,17} Even moderate elevations in maternal glucose within the physiological range have been linked to placental remodeling, highlighting the sensitivity of this organ to metabolic shifts.¹⁸

Abnormal placental invasion, as seen in placenta accreta spectrum, represents another major pathology where defective decidualization and excessive trophoblast invasion lead to life-threatening hemorrhage at delivery.¹⁹

Altogether, these disease settings highlight that the placenta is a central determinant of pregnancy outcome. Whether through impaired angiogenesis, metabolic dysregulation, infectious injury, or abnormal invasion, placental dysfunction triggers a cascade of consequences. The spectrum of outcomes ranges from preterm birth and stillbirth to fetal growth restriction, organ malformations, and developmental deficits in various fetal organs.

THE INFLUENCE OF THE PLACENTA ON FETAL HEART DEVELOPMENT

The heart is the first organ to form in embryogenesis, beginning development by the end of the second week of pregnancy. Disruption in this process can lead to the formation of structural or functional abnormalities,

Highlights

- Pregnancy disorders disrupt placental biology, thereby driving adverse fetal cardiovascular outcomes.
- The placenta-fetal heart axis plays a critical role in cardiac development, with dysfunction increasing susceptibility to congenital heart disease.
- Impaired placental function contributes to maternal cardiac maladaptation and heightened cardiovascular stress responses.
- Abnormal placentation shapes long-term cardiovascular health in both mother and offspring.

many of which arise in the early gestation period and can culminate in congenital heart diseases (CHDs). CHDs are a group of conditions that refer to structural or functional abnormalities of the heart and greater vessels that are present at birth. They are the most prevalent congenital disorder, estimated to be present in 9 per 1000 live births, and account for $\approx 35\%$ of infant deaths.²⁰ In addition, CHDs are associated with perinatal morbidity and mortality, being identified in $\approx 11\%$ of still births.²¹ There are known long-term consequences of CHDs resulting in increased morbidity risks with adult survivors developing cardiovascular disease and cancer.²²

During the first trimester, placental and cardiac development proceed in parallel and are biologically interdependent, with common genetic and molecular pathways, such as signaling pathways involving the micronutrient folate via Wnt/ β -catenin, GATA factors, and HAND1, being identified in early development of both organs.^{23–25} The fetal heart is the first organ formed during gestation; however, its morphogenesis occurs within an embryonic milieu that is actively regulated by early placentation. Notably, early placental structures maintain a physiologically low-oxygen environment through restricted maternal blood flow into the intervillous space, a condition essential for normal cardiac development²⁶ (Figure 1). Hypoxia-sensitive pathways, including hypoxia-inducible factor signaling, influence myocardial proliferation, cardiac looping, and chamber specification during these early stages.²⁷ As placental villous branching and vascularization progress across the first trimester, gradual increases in maternal perfusion alter fetal oxygenation and hemodynamic load at critical windows of cardiac development.²⁶ These changes coincide with key processes such as septation, valve formation, and coronary angiogenesis, all of which are sensitive to shear stress, blood flow patterns, and metabolic cues.²⁸ At the completion of the first trimester, the heart is fully formed in its basic structure, and the placenta becomes the dominant metabolic and respiratory organ, facilitating further organogenesis.

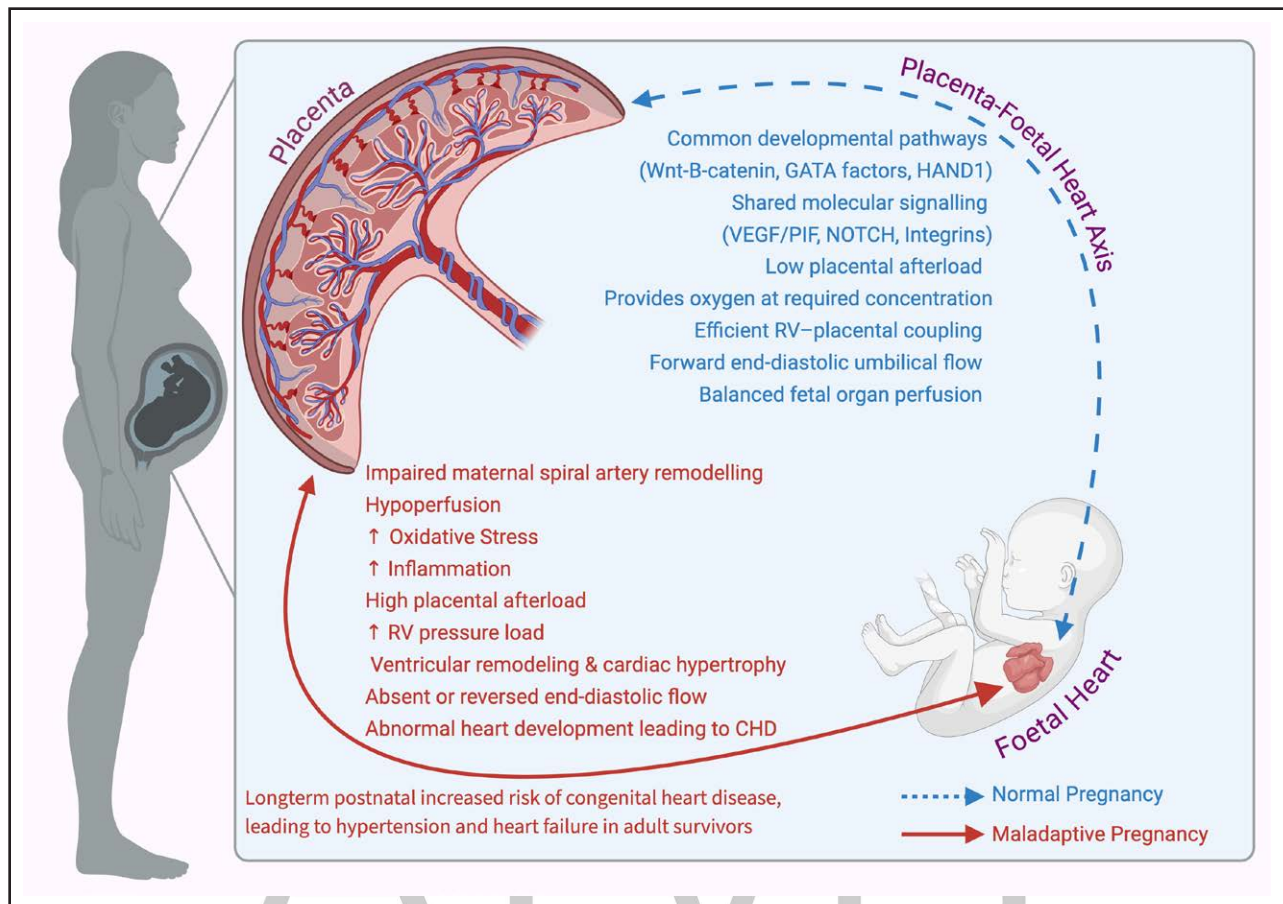


Figure 1. Schematic illustration of fetal cardiac adaptation in normal pregnancy and maladaptation resulting from placental dysfunction.

CHD indicates congenital heart disease; and VEGF, vascular endothelial growth factor.

Arteriosclerosis, Thrombosis, and Vascular Biology

Experimental models and human observational studies increasingly support the presence of the fetal heart-placenta axis, whereby abnormalities in early placental development can disrupt embryonic cardiac organogenesis, thereby increasing susceptibility to CHDs. Disruptions to the essential invasion of trophoblasts and remodeling of maternal spiral arteries result in impaired maternal-fetal oxygen transfer, adversely impacting fetal cardiac development. Clinical studies have associated placental dysfunction with an increased incidence of CHDs.²⁹ A study investigating the presence of placental defects in murine models found that in early lethality (before embryonic day 14.5), placenta malformation was present in the majority of models, and from mid-gestation onward, 68% of knockout lines showed placenta dysmorphologies.³⁰ Furthermore, Perez-Garcia et al³⁰ identified that the placenta defects correlated with abnormalities in vascular and cardiac development. Molecular pathways that govern both placental and cardiac development, including VEGF (vascular endothelial growth factor)/PIGF, NOTCH, and integrins, appear disrupted when placentalation fails, suggesting a direct developmental link between the 2 organs.³¹

Placenta dysfunction and cardiac malformations frequently coexist in complicated pregnancies. In preeclampsia, the hypoperfused placenta and oxidative stress induce inflammation via proinflammatory cytokine release and endothelial activation.³² This dysfunctional placental environment impacts fetal hemodynamics and the development of the heart.³³ Doppler and fetal echocardiographic studies demonstrate increased myocardial performance index, decreased systolic and diastolic functions, and altered ventricular structure in fetuses exposed to preeclampsia.³⁴ These cardiac changes may not fully resolve after birth, with studies showing an increased risk of offspring developing a range of CHD,³⁵ including septal, valve, and pulmonary defects.³⁵ During the neonatal period, offspring from preeclamptic pregnancies exhibit reduced left ventricular (LV) ejection fractions, myocardial noncompaction, and increased pulmonary vascular resistance.³⁶ These early cardiac abnormalities are followed by a heightened risk of cardiovascular disease later in life.³⁷

A common comorbidity of preeclampsia is IUGR, occurring in combination or independently of preeclampsia. IUGR has been associated with an increased

incidence of CHD³⁸ due to overlapping mechanisms of poor formation of the placenta vascular network, leading to oxidative stress.³⁸ Cardiovascular remodeling associated with hypertrophy, ventricular and atrial dilation, and myocardial structural abnormalities is present in IUGR.³⁹ In clinical settings, smaller placenta weight has been associated with IUGR and the formation of certain CHD, namely, tetralogy of Fallot, double-outlet right ventricle, major ventricular septal defect, and transposition of the great arteries.⁴⁰

Another risk factor for CHD linked to placental dysfunction is the presence of maternal metabolic syndromes, such as obesity and diabetes. Epidemiological evidence shows that maternal obesity is associated with increased incidence of CHD, notably ventricular outflow tract defects, aortic arch defects, and transposition of the great arteries. Furthermore, studies observed a dose-dependent relationship between maternal body mass index and the risk of CHD, stillbirths, and postnatal fatalities.^{41,42} Obese pregnancies present with metabolic dysregulation, including hyperglycemia, insulin resistance, dyslipidemia, and oxidative stress.^{43,44} These metabolic disturbances are accompanied by endothelial dysfunction and inflammation.⁴⁵ Insufficient placenta vascularization in obese pregnancies alters nutrient and oxygen delivery, adversely affecting fetal vascular and cardiac development, providing a link between maternal metabolic disease, placental pathology, and increased CHD risk in the offspring.

Gestational diabetes causes a 4-fold increased risk of CHD in offspring, rising to 8-fold when acute diabetic complications are present.⁴⁶ with studies reporting atrial and ventricular defects, and tetralogy of Fallot is prevalent among these populations.⁴⁷ Hyperglycemia is a hallmark of diabetes and has been established as a teratogenic mechanism underlying CHD risk. Glucose is transported from the maternal circulation to the fetus via facilitated diffusion across the placenta, a tightly regulated process that is essential in organogenesis. Under hyperglycemic conditions, excessive fetal glucose exposure disrupts metabolic and signaling pathways; this dysregulation can lead to cardiac malformations.⁴⁸

Collectively, substantial evidence supports a central role for placental development and function in determining pregnancy outcomes and fetal cardiovascular health. The placenta and fetal heart engage in a dynamic, bidirectional relationship throughout gestation; the placenta-fetal heart axis is essential for both coordinated organ development and the progression of organogenesis. Complicated pregnancies with placental dysfunction have an increased risk of offspring developing CHD, providing the hypothesis that effective prevention or mitigation of CHD may require therapeutic strategies that target placental function and health, in addition to fetal cardiac development itself.

MATERNAL CARDIAC REMODELING IN HEALTHY PREGNANCY

Maternal cardiac remodeling in healthy pregnancy follows a structured, trimester-dependent progression that reflects coordinated hemodynamic and myocardial adaptation. Pregnancy imposes a heavy hemodynamic burden on the maternal heart and vasculature. Under healthy conditions, maternal blood volume expands by 30% to 50%, systemic vascular resistance declines, and cardiac output rises (through increased stroke volume and modest tachycardia) to support the demands of both placenta and fetus.⁴⁹ In the first trimester, systemic vasodilation reduces afterload and mean arterial pressure, while stroke volume and cardiac output begin to rise as early hemodynamic adaptation is established. In a large prospective study of 559 nulliparous women assessed serially during pregnancy and at 1 year postpartum, mean arterial pressure and total vascular resistance declined significantly in early gestation, accompanied by increases in stroke volume and cardiac index, demonstrating early cardiovascular compensation.⁵⁰ Similarly, trimester-specific echocardiographic data from a cohort of 412 healthy pregnant women showed that cardiac output increases are evident from early pregnancy and progress across gestation, supporting a structured pattern of cardiovascular adaptation.⁵¹ At this stage, structural cardiac changes are modest, with preserved chamber geometry and systolic function, indicating that early adaptation is primarily hemodynamic and compensatory rather than structural. These early vascular and hemodynamic adjustments are considered essential for accommodating placental development while preventing excessive maternal cardiac workload.⁵¹

As pregnancy progresses into the second trimester, sustained plasma volume expansion increases preload, providing a direct stimulus for structural cardiac remodeling. Stroke volume and cardiac index continue to rise, while systemic vascular resistance remains reduced.⁵⁰ Arterial stiffness reaches its lowest level in mid-pregnancy before gradually increasing toward term, reflecting dynamic vascular adaptation.⁵² Under these loading conditions, longitudinal echocardiographic studies demonstrate increases in LV end-diastolic and end-systolic volumes, together with preserved ejection fraction.⁵¹ Earlier research similarly documented enlargement of LV dimensions and increased LV mass during this period.⁵³ These findings are consistent with physiological eccentric remodeling, characterized by chamber dilation with maintained systolic performance in response to sustained volume load. In the third trimester, continued hemodynamic demand further accentuates myocardial adaptation. Chamber enlargement and elevated cardiac output persist under sustained preload. Four-dimensional echocardiographic evaluation in healthy late pregnancy shows increased LV volumes and cardiac output, with

a mild but significant reduction in ejection fraction that remains within physiological limits.⁵⁴ Advanced speckle-tracking imaging has demonstrated a progressive reduction in LV global longitudinal strain accompanied by increased LV twist, despite preserved conventional ejection fraction.⁵⁵ These changes reflect altered myocardial mechanics and compensatory torsional augmentation rather than intrinsic systolic dysfunction. In a subset of women, mild diastolic alterations emerge toward term, consistent with transient hemodynamic stress rather than structural disease.⁵⁰ Complementary cardiovascular magnetic resonance imaging confirms significant increases in LV end-diastolic volume and LV mass in late pregnancy, further supporting the concept of adaptive eccentric remodeling.⁵⁶

Importantly, these trimester-dependent structural and functional changes are largely reversible postpartum. Serial imaging demonstrates regression of chamber enlargement and normalization of hemodynamic parameters within months after delivery, underscoring the plasticity of the maternal myocardium.⁵⁷ Overall, maternal cardiac remodeling in healthy pregnancy is structured, adaptive, and functionally preserved across gestation.

MATERNAL CARDIAC MALADAPTATION AS A CONSEQUENCE OF PLACENTAL DYSFUNCTION

The placenta-maternal heart axis describes the bidirectional relationship between maternal cardiovascular function and placental development (Figure 2). Emerging data indicate that maternal cardiovascular maladaptation may precede overt clinical manifestations of placental disease, and vice versa. Inadequate trophoblast invasion and failed remodeling of spiral arteries create a high-resistance uteroplacental circulation, which elevates maternal afterload and impairs hemodynamic adaptation.⁵⁸ This placental pathology initiates a maladaptive cycle in which compromised uteroplacental blood flow and maternal cardiovascular burden are mutually aggravating. Clinical and imaging studies provide strong support for this mechanistic link. A decline in maternal cardiac output and abnormal umbilical artery Doppler findings suggested that placental dysfunction contributes to maternal cardiac maladaptations, where impaired placental perfusion increased cardiac strain and predisposed to adverse neonatal outcomes.⁵⁹ Mid-gestation echocardiographic

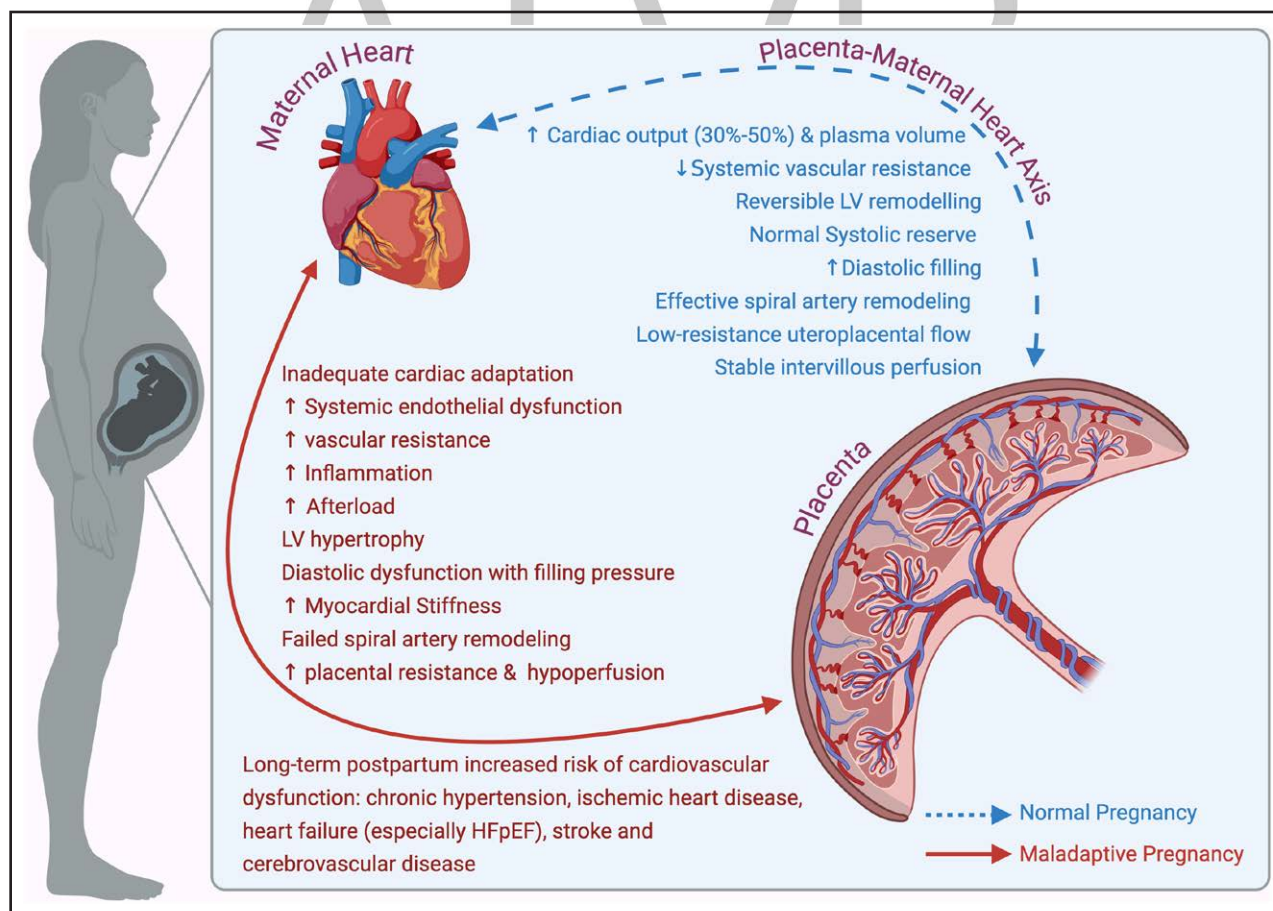


Figure 2. Schematic illustration of maternal cardiac adaptation in normal pregnancy and maladaptation resulting from placental dysfunction.

LV indicates left ventricular.

studies show that women who later develop preeclampsia exhibit higher LV wall stress, reduced compliance, and elevated systemic vascular resistance compared with normotensive pregnancies.⁶⁰ These subclinical findings indeed support the bidirectional pathogenic interaction between maternal heart and placenta.

Preeclampsia represents one of the clearest examples of maladaptive placenta-heart interactions. It has been shown to be associated with impaired maternal cardiovascular adaptation during pregnancy, evidenced by subclinical myocardial dysfunction and structural cardiac changes that correlated with elevated NT-proBNP (N-terminal pro-B-type natriuretic peptide) levels and earlier delivery.⁶¹ Longitudinal studies indicate that these abnormalities often persist postpartum, conferring a higher lifetime risk of hypertension, ischemic disease, and heart failure with preserved ejection fraction.⁶² Impaired placental development in preeclampsia has been associated with higher postpartum cardiovascular disease risk, indicating that inadequate maternal cardiovascular adaptation during pregnancy may have lasting cardiovascular consequences.⁶³ Furthermore, maternal cardiac maladaptation has been linked to placental dysfunction in preeclampsia, driven by placental cellular senescence, with senescence-associated factors such as activin A contributing to impaired maternal cardiac function.⁶⁴ A large cohort study found that postpartum preeclampsia, though less frequent than its antepartum form, was strongly associated with placental abruption, preterm birth, and severe maternal morbidity, reflecting significant cardiovascular stress.⁶⁵ These findings suggest that postpartum preeclampsia may originate from antenatal placental dysfunction, leading to persistent maternal vascular and cardiac maladaptations, including endothelial injury, elevated afterload, and impaired cardiac remodeling that extend beyond delivery and heighten the mother's long-term risk of cardiovascular disease.⁶⁵ Such findings support the view that preeclampsia represents a cardiovascular stress test that unmasks susceptibility to chronic disease. Our recent review highlights that immune dysregulation in preeclampsia leads to placental vascular injury and systemic endothelial dysfunction, which elevates maternal vascular resistance and hemodynamic stress.⁸ These changes impair normal cardiac adaptation, causing the maternal heart to undergo maladaptations such as increased afterload, hypertrophy, and diastolic dysfunction, thereby linking placental immune injury to maternal cardiovascular strain.⁸

The cardiovascular implications of placental pathology extend beyond preeclampsia. Placental abruption, characterized by premature detachment of the placenta, causes acute hemodynamic instability, sympathetic activation, and, sometimes, hypovolemic stress.⁶⁶ The abrupt circulatory changes and endothelial injury place intense pressure on the maternal myocardium, occasionally resulting in transient or chronic cardiac impairment.⁶⁷

Another important complication of the placenta is placenta accreta spectrum, a pregnancy disorder in which the placenta abnormally adheres to or invades the uterine wall, resulting in a high risk of severe hemorrhage and complex obstetric management.¹⁹ It is marked by excessive trophoblast invasion and may cause chronic inflammation, abnormal vascular connections, and massive hemorrhage at delivery, all of which exacerbate cardiac workload and compromise myocardial performance.⁶⁸ Consistent with this, postpartum hemorrhage, which is frequent in placenta accreta spectrum, has been linked with increased long-term cardiovascular risk, suggesting that severe hemodynamic stress during delivery may have lasting effects on maternal cardiovascular health.⁶⁹

These findings indicate that placental disorders are not isolated obstetric events but reflect systemic vascular and cardiac vulnerability. Beyond the peripartum period, women with prior placental disorders face an elevated risk of chronic hypertension, ischemic heart disease, and heart failure, suggesting that these conditions reveal an underlying predisposition to cardiovascular disease.^{70,71} Understanding how placental dysfunction alters maternal hemodynamics and cardiac remodeling is crucial for improving prediction, prevention, and long-term management of maternal cardiovascular risk.

LONG-TERM IMPACT OF PLACENTATION DEFECTS ON MATERNAL AND FETAL HEART

Emerging evidence highlights that defective placentation, independent of etiology, profoundly impacts the cardiovascular system in both the mother and the fetus. In a cohort of >4000 women between 19 and 23 weeks of gestation, reduced PlGF and elevated sFlt-1 were tied to higher maternal blood pressure and slight alterations in heart structure and function, suggesting that suboptimal placental angiogenesis contributes to mild maternal cardiac remodeling during mid-pregnancy.⁷² Defective placentation, as seen in preeclampsia or fetal growth restriction, has been associated with a significantly increased long-term risk of maternal cardiovascular disease, particularly hypertension, despite minimal structural cardiac changes, decades after pregnancy.⁷³ Advanced echocardiographic and magnetic resonance imaging studies reveal that women with preeclampsia often exhibit impaired diastolic function, altered ventricular-arterial coupling, and myocardial remodeling that may persist postpartum.^{74,75} Epidemiological evidence underscores that women with a history of preeclampsia carry a 2- to 4-fold higher lifetime risk of ischemic heart disease, stroke, and heart failure, supporting the notion that placental pathology is both a marker and mediator of future cardiovascular disease.⁷⁶ These risks are amplified

when placental insufficiency occurs in the context of maternal obesity, insulin resistance, or gestational diabetes, conditions that augment placental inflammation, impair mitochondrial function, and heighten cardiovascular strain.^{77–79} It has been shown that women who experienced retained placenta with hemorrhage had an increased risk of long-term cardiovascular disease.⁸⁰ It is clear that inadequate placentation is not confined to pregnancy morbidity but sets the stage for long-term maternal cardiovascular vulnerability, yet most women remain unaware of this risk, underscoring the need for education and preventive care.⁸¹

For the fetus, placental insufficiency initiates a distinct but equally significant cascade of cardiovascular alterations. Restriction of oxygen and nutrient supply due to placental defects alters preload and afterload, driving cardiac remodeling in utero. In addition, longitudinal studies indicate that individuals born with growth restriction display altered LV geometry, impaired vascular function, and heightened cardiometabolic risk in adolescence and adulthood, clearly underscoring the enduring impact of adverse intrauterine hemodynamics.⁸² It is evident that defective placentation compromises fetal cardiovascular development, producing effects that persist beyond gestation and impact lifelong cardiac health.

CONCLUSIONS AND FUTURE DIRECTIONS

The placenta is a powerful organ that not only impacts the development of the fetal heart but can also have long-term consequences on the cardiac health of both mother and child.

One of the key challenges that face research into how placental dysfunction affects fetal and maternal cardiac health is determining when exactly these changes arise. Recent advances in imaging techniques have enabled the early detection of fetal CHD within the first trimester between 12 weeks and 0 days, and 13 weeks and 6 days. Targeted imaging of the cardiac position and axis, the 4-chamber view of the heart, and imaging the vessels and trachea allow for early diagnosis of fetal CHD. However, at present, this first-trimester imaging is reserved for high-risk groups (including maternal diabetes, hereditary CHD history, or other fetal anomalies).⁸³ In pregnancies considered low risk, CHDs are not definitively diagnosed until the 20th week of pregnancy, by which time the placenta and the heart are fully developed.

Thus, while it is still vital to continue research into the mechanisms by which placental dysfunction can cause poor cardiac outcomes, parallel studies are much needed to identify novel biomarkers early in pregnancy that can predict adverse cardiac outcomes. Such biomarkers

could greatly improve patient care and offer potential therapeutic strategies that target pathways involved in dysregulation of the placenta-heart axis.

ARTICLE INFORMATION

Affiliation

Centre for Biochemical Pharmacology, William Harvey Research Institute, Queen Mary University of London, United Kingdom.

Author Contributions

M. Rana, U.R. Wilcox, and S. Nadkarni wrote the manuscript. U.R. Wilcox designed Figures 1 and 2. Figures were created in BioRender.

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